README

Curve25519-family algebraic track beyond t = 2: results (2026-11-02)

Protocol: PROTOCOL.md (frozen; Amendment 1 adds hybrid guessing, Amendment 2 replaces the end-to-end ladder by a stage measurement plus end-to-end checks, both written before their data). PDF: REPORT.pdf. Code: `pkm-semaev` membership forms isogeny / bits (#3284), `guess=G` (#3287), harness `curve25519_family f4cost` and `support` (#3288, #3290).

Class: stage measurement of the algebraic track plus two end-to-end checks. It is the algebraic track's distance to the generic baseline. No improvement claim. Every number below is for **k = 20, m = 2**, one miniature curve.

Q-A: support of membership models (exact counts)

`analysis/support.jsonl`, `analysis/support-edwards.jsonl`. At width 2^10:

membership model	signed classes	λ = M/ℓ
Weierstrass x in [0, 2^t)	458	0.959
Montgomery x in [0, 2^t)	533	0.987
Edwards y in [0, 2^t)	499	0.979
T_0-invariant w = x + 1/x in [0, 2^t)	268	0.669
T_0-invariant Edwards y^2 in [0, 2^t)	245	0.604
PKM isogeny tower, height t	498	0.978
PKM Kummer tower	exists only for t ≤ 2 (p ≡ 5 mod 8)	

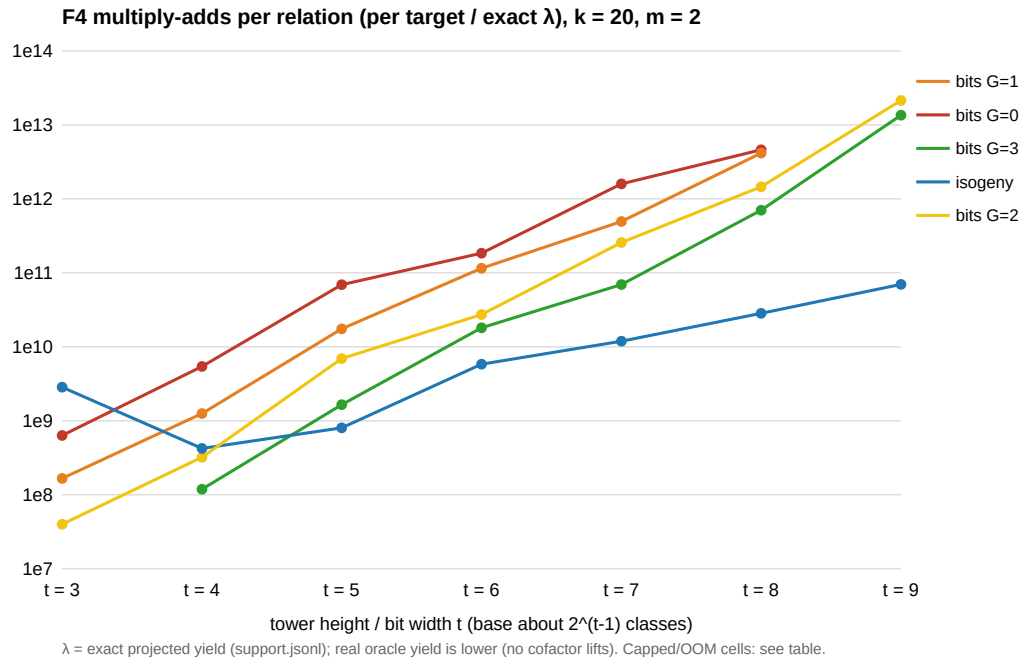
The table is limited to these intervals. It measures one window, [0, 2^t), per coordinate, on one curve, and only its support. It does **not** show that Edwards or invariant coordinates bring no advantage to relation solving. Other subsets in those coordinates (offsets, other windows, sets structured in y) and the algebraic cost of S_{m+1} in them are untested. Lead 5 is testing the latter. The statement in commit `d3adb94a39` that the Edwards interval "carries the same support" means only this measured interval.

Q-D: F4 cost per target and per relation

`analysis/f4cost.jsonl` (200 random targets per cell, or fewer where the 600 s cap stopped it, marked "cap"). Per relation = F4 multiply-adds per target ÷ λ (the exact projected yield). This is a lower bound on the oracle's real cost per relation: the oracle finds only exact decompositions of R, not of R + K for K ∈ E[8] (lead 2). "per hit" uses the hits actually observed (1–7 per cell; large uncertainty).

form	t	G	columns	targets	F4 mul-adds / target	solving degree	hits	per hit	λ	per λ-relation
bits	3	0	3	200	8.70e+04	5	0	—	0.0001	6.34e+08
bits	3	1	3	200	2.29e+04	5	0	—	0.0001	1.67e+08
bits	3	2	3	200	5.46e+03	4	0	—	0.0001	3.98e+07
isogeny	3	0	1	200	4.34e+04	5	0	—	0.0000	2.85e+09
bits	4	0	7	200	4.05e+06	6	0	—	0.0007	5.42e+09
bits	4	1	7	200	9.36e+05	5	0	—	0.0007	1.25e+09
bits	4	2	7	200	2.40e+05	5	0	—	0.0007	3.21e+08
bits	4	3	7	200	8.88e+04	4	0	—	0.0007	1.19e+08

form	t	G	columns	targets	F4 mul-adds / target	solving degree	hits	per hit	λ	per λ-relation
isogeny	4	0	9	200	5.23e+05	5	0	—	0.0012	4.23e+08
bits	5	0	11	200	1.27e+08	6	0	—	0.0018	6.90e+10
bits	5	1	11	200	3.23e+07	6	0	—	0.0018	1.75e+10
bits	5	2	11	200	1.28e+07	5	0	—	0.0018	6.93e+09
bits	5	3	11	200	3.04e+06	5	0	—	0.0018	1.65e+09
isogeny	5	0	18	200	3.96e+06	5	0	—	0.0049	8.01e+08
bits	6	0	25	200	1.76e+09	7	1	3.51e+11	0.0095	1.84e+11
bits	6	1	25	200	1.10e+09	6	1	2.20e+11	0.0095	1.15e+11
bits	6	2	25	200	2.61e+08	6	1	5.21e+10	0.0095	2.74e+10
bits	6	3	25	200	1.72e+08	6	1	3.44e+10	0.0095	1.81e+10
isogeny	6	0	25	200	5.55e+07	5	1	1.11e+10	0.0095	5.83e+09
bits	7	0	58	63 (cap)	7.94e+10	7	0	—	0.0499	1.59e+12
bits	7	1	58	164 (cap)	2.47e+10	7	1	4.04e+12	0.0499	4.94e+11
bits	7	2	58	200	1.28e+10	7	2	1.28e+12	0.0499	2.57e+11
bits	7	3	58	200	3.47e+09	6	2	3.47e+11	0.0499	6.95e+10
isogeny	7	0	65	200	7.46e+08	5	4	3.73e+10	0.0627	1.19e+10
bits	8	0	114	6 (cap)	8.29e+11	8	0	—	0.1795	4.62e+12
bits	8	1	114	6 (cap)	7.47e+11	8	0	—	0.1795	4.16e+12
bits	8	2	114	12 (cap)	2.61e+11	7	0	—	0.1795	1.45e+12
bits	8	3	114	24 (cap)	1.26e+11	7	0	—	0.1795	7.02e+11
isogeny	8	0	136	200	6.98e+09	5	7	1.99e+11	0.2461	2.83e+10
bits	9	2	233	1 (cap)	1.21e+13	8	0	—	0.5622	2.15e+13
bits	9	3	233	1 (cap)	7.62e+12	8	0	—	0.5622	1.35e+13
isogeny	9	0	256	77 (cap)	4.39e+10	5	5	6.76e+11	0.6277	7.00e+10
bits	9	0	233	OOM	killed at ~4.5 GB RSS (shared memory cgroup)					
bits	9	1	233	OOM	killed at ~4.5 GB RSS					



per-relation cost

Reading, at this scope: - **Hybrid guessing works on the bits form.** Per-target F4 work falls 10–46× from G = 0 to G = 3 at t = 4–7 (t = 4: 46×, 5: 42×, 6: 10×, 7: 23×), and is still falling at G = 3. Prediction C1 (optimum at G = 1–3) is not confirmed: the optimum lies at G ≥ 3. Lead 4 extends this. - **The isogeny tower scales best with t.** Its per-λ-relation cost grows 2–7× per step in t (t = 4–9), against 4–19× for bits at G = 3. Its solving degree stays 5 for t = 3–9, while bits reaches 8. - **The cheapest cells per λ-relation are the smallest bases:** bits t = 3, G = 2 ($4.0 \cdot 10^7$), then bits t = 4, G = 3 ($1.2 \cdot 10^8$). The isogeny t = 3 value rests on a one-class tower. - Predictions B1 (both new forms run beyond t = 2) **held**. B3 (bits within 4× of isogeny at equal t) **failed** from t ≥ 5: unguessed bits costs 32× (t = 5, 6) to 106–119× (t = 7, 8) the isogeny form per target.

End-to-end checks (ecbench, verified, replay-audited)

`sessions/e2e-iso-t6-k20`, `sessions/e2e-bits-t3-g2-k20` (2 public targets, 1 round each, frozen binary, L0/L1 counts only; the generic arm is the current k = 20 baseline from #3290).

arm	S, w = 0	S, w = 0.1	S, w = 1
`rho.negation`	1.71	1.84	2.99
generic baseline `ksum:m`=3, N = 11	7.92	8.43	13.0
algebraic, isogeny t = 6	42.4	$1.9 \cdot 10^8$	$1.9 \cdot 10^9$
algebraic, bits t = 3, G = 2	795	$2.5 \cdot 10^6$	$2.5 \cdot 10^7$

The stage ranking carries over where F4 is priced. The protocol's cheapest cell, bits t = 3, G = 2, is about 80× cheaper than isogeny t = 6 at w ≥ 0.1, and about 19× worse at w = 0, because a 3-column base needs many more walk steps. The best measured algebraic configuration is about **$3 \cdot 10^5$×** the generic baseline at w = 0.1 on this curve. That is the algebraic track's distance, on 2 targets, at k = 20 only.

Problem record

`problems/curve25519-family-algebraic-ic-tower.md`: its "missing capability" (an isogeny-tower Semaev oracle) is now provided by #3284. The obstruction for useful bases is now cost, not existence.

Next

Leads 2–5 are being completed one at a time (exclusive build slot): cofactor lifts (2), m = 3 chained S_3 (3), deeper guessing incl. isogeny guessing (4), Edwards-coordinate S_3 (5).

PROTOCOL

Curve25519-family algebraic track beyond t = 2 — protocol

Frozen 2026-11-02 before any data in this directory. It follows [problems/curve25519-family-algebraic-ic-tower.md](): `pkm-semaev` accepted only the Kummer tower `g·μ_(2^t)`, which needs `2^t | p − 1`. Curve25519-family primes have `p ≡ 5 (mod 8)`, so `t ≤ 2`.

What changes in the code

`pkm-semaev` gains a membership layer (`Membership`):

- **kummer** (unchanged): `x = g·y_0`, `y_j² = y_(j+1)`, `y_(t−1)² = 1`.
- **isogeny** (PKM's isogeny form, any p): `x = y_0`, and each Vélu step `y_(j+1) = y_j + τ_j/(y_j − ξ_j)` is cleared to `y_j² = y_j·y_(j+1) − ξ_j·y_(j+1) + ξ_j·y_j − τ_j`. The top level is `y_(t−1)² = (c + ξ)·y_(t−1) − cξ − τ`. Base `prime-isogeny:t=T`.
- **bits** (the owner's interval encoding, any p): `x = lo + Σ 2^j b_j`, `b_j² = b_j`, on any base whose abscissae lie in `[lo, lo + 2^T)`, with oracle parameter `bits=T`.

The F4 engine (`f4_fp_tower`) already takes all three rule shapes. The grid reference and the lift are unchanged. New unit tests check, for both new forms, that F4 equals the grid and the exact decomposable set on random and planted targets, and that every relation sums to its target.

Q-A: quotient support (exact, `curve25519_family support`)

For each form and t, on the k = 20 and k = 24 miniatures, report: physical points; classes after torsion compression (`[8]P`); classes after signed compression; `M` (distinct projected signed 2-sums); `λ = M/ℓ`; and the naive count taking every physical pair (its overstatement factor). Forms: `xw` (Weierstrass interval), `xm` (Montgomery interval), `w` (the T_0-invariant `x + 1/x` interval, i.e. the 2-isogenous curve's coordinate), `isogeny`, and `kummer` where it exists. t = 2 … 10.

Predictions: - **A1.** Interval forms have about `2^(t−1)` signed classes. The `w` form has about the same number of classes per interval width as `xm`, but twice the physical points per class. That is the 2:1 T_0 compression, so it costs about 2× in physical points and nothing in classes. - **A2.** The naive pair count overstates `M` by about 8 (the cofactor) times the per-class multiplicity.

Q-B: end-to-end ecbench ladder (algebraic track, m = 2, mode=full)

k = 20 miniature, 4 public targets, 1 round, target seed `2026110700`, timeout 1800 s per run. Arms: - `iso-t{3..9}`: `prime-isogeny:t=T` + `pkm-semaev:m=2,mode=full`; - `bits-t{3..9}`: `prime-abscissa:size=N_T` + `pkm-semaev:m=2,bits=T`, where `N_T` = the number of abscissae in `[0, 2^T)` carrying a point (from Q-A, form `xw`); - `kummer-t2`: the previous ceiling; - `inc`: the generic incumbent; `rho-neg`: matched rho.

Reported per arm: verified runs; S at w = 0 / 0.1 / 1 (`ecbench band`); F4 multiply-adds per call; solving degree; relations and trials. A run that times out is reported, not dropped.

Predictions: - **B1.** Both new forms run and verify at t > 2. - **B2.** S at w = 0 falls with t up to the point where trials per relation stop dominating. S at w ≥ 0.1 is dominated by F4 multiply-adds, which grow with t. No algebraic arm comes within 10× of `inc` at w = 0.1. - **B3.** At equal t, bits uses as many tower variables as isogeny (t per summand). Its S_3 has degree 4 in the bits against degree 4 in `y_0`, so their F4 costs are within 4× of each other.

These are measurements of the algebraic track's distance, not improvement claims against the generic incumbent. Hybrid guessing, chained S_3 and shared preprocessing (owner's item 6) are next steps, and only once Q-B shows which model is cheapest.

Amendment 1 (2026-11-02, before any Q-B or Q-C data): hybrid guessing

Owner request: add hybrid guessing to the bits form. `pkm-semaev` gains `guess=G` (bits form only, `G < bits`, $G \le 12$). For a target, the oracle enumerates the `2^G` assignments of the top G bits of the first summand, runs F4 on each reduced system, and stops at the first consistent branch. The target is refuted only when all branches refute. Every branch's F4 multiply-adds are charged; `pkm_guess_branches_solved` counts the systems solved. Unit test: for `G ∈ {1, 2}` the verdicts equal the grid's and the exact decomposable set on random and planted targets.

Q-C: guessing sweep

The bits form on the k = 20 miniature at the t that the Q-B ladder shows cheapest in S at w = 0.1, with `G ∈ {0, 1, 2, 3, 4}` at that t. The same 4 targets, seed and timeout as Q-B. The cost target is the owner's `C(G) = 2^G·C_solve(after G)` plus grid recovery and verification, read as F4 multiply-adds per call (`ecbench` solver block) and as S across the band.

Prediction **C1**: total F4 multiply-adds per target first fall with G, because each branch has G fewer Boolean variables and a lower solving degree. They then rise once `2^G` dominates. The best G lies in 1 … 3. The relation verdicts are independent of G, since refutation is exact.

Amendment 2 (2026-11-02, after the timing smoke, before any Q-B, Q-C or Q-D data)

The timing smoke (`sessions/smoke-k20`, one target, `interrupted` after 4 of 8 runs; used for sizing only) showed: - `iso-t3`: 131,202 F4 calls at 43k multiply-adds each. It exhausted its 10^8-trial cap with no relation set (1 class, matching Q-A). - `bits-t3`: 85,005 calls at 87k each for 2 relations, S = 378. - `iso-t6`: no finish within 30 minutes on one target.

At these costs the frozen Q-B ladder (14 arms × 4 targets, 1800 s timeouts) would take days. Q-B and Q-C are therefore replaced by a stage measurement with the same instrument (the `pkm-semaev` oracle, same code path, called directly) plus the exact Q-A yields:

Q-D: F4 cost per call and per relation

`curve25519_family f4cost <k> <form> <t> <G> <targets> <seed>`, k = 20, forms `bits` (with `G ∈ {0, 1, 2, 3}` where `G < t`) and `isogeny`, t = 3 … 9. 200 random subgroup targets per cell, seed `2026110800 + t`, and a per-cell wall cap of 600 s; a capped cell reports how many targets it finished. Reported per cell: - F4 calls, multiply-adds per target, the maximum solving degree, and undecided systems; - the hit rate, against the exact Q-A λ; - the projected multiply-adds per relation, `(multiply-adds per target)/λ`.

One verified end-to-end ecbench run then measures the cheapest cell, to confirm that the stage figure carries over to S. Predictions: - **B2** stands as stated. - **C1** is restated per target: total multiply-adds per target are minimised at $G \in \{1, 2, 3\}$.